Summary: Design of a Magnetic Marble Run Exhibit for the Discovery Space Museum

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Science, Technology, Engineering, and Mathematics (STEM) are subjects that elevate students' capabilities in areas that require problem-solving [1]. As a way to help enhance students STEM skills, a junior-level design course at Pennsylvania State University has tasked our team with designing a magnetic marble course that allows students the creativity to create their own course as shown in Figure 1. As stated in the project requirements [2], the magnetic marble run is to not exceed two feet in any dimension while stored and the exhibit should have at least one additively manufactured part. Moreover, to satisfy the immediate purpose of the exhibit, the exhibit must simultaneously engage, entertain, and educate the children about physical or scientific principles [2]. This project summary presents the design of our magnetic marble run track.

To come up with the design concepts for the magnetic marble run's structural main frame and its additively manufactured parts, our team first created a functional decomposition chart. This chart allowed us to understand the function of the exhibit needed to effectively complete a marble run. By using the project requirements [2], our team identified 4 main customer needs: is durable, is engaging, is aesthetic, is safe. After assigning metrics and specifications to match each of those needs, our team then proceeded to generate three concepts for each of the subsystems. When rating them using a concept scoring matrix, our team ultimately decided on one concept for the face of frame, for the structural frame, for the component connection, and one concept for the randomization element. As shown in Figure 1, the main part features a magnetic surface, with an inclined frame, and a plinko board that help for randomization. Figure 2a features an additively manufactured component that has slots for magnets to be embedded.

Throughout the last section of the project, our team build and tested a series of prototypes: alpha 1, alpha 2, beta and a final product. The Final product can be seen in figure 2b. Throughout the different prototypes of our main product, we also created different 3D additive manufacturable components that add dynamic to the course and allow the user to create more varieties of tracks. Included in these designs were different sloped ramps, curves, and straight lines. On December 1st, our team created a short video of running marbles through the exhibit and explaining the scientific concepts and design choices of the exhibit [3].

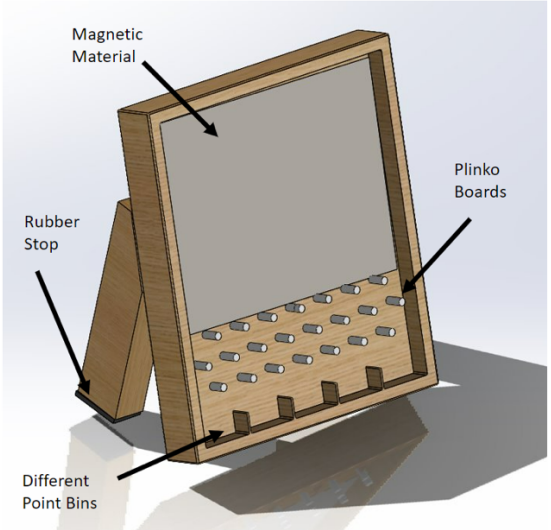
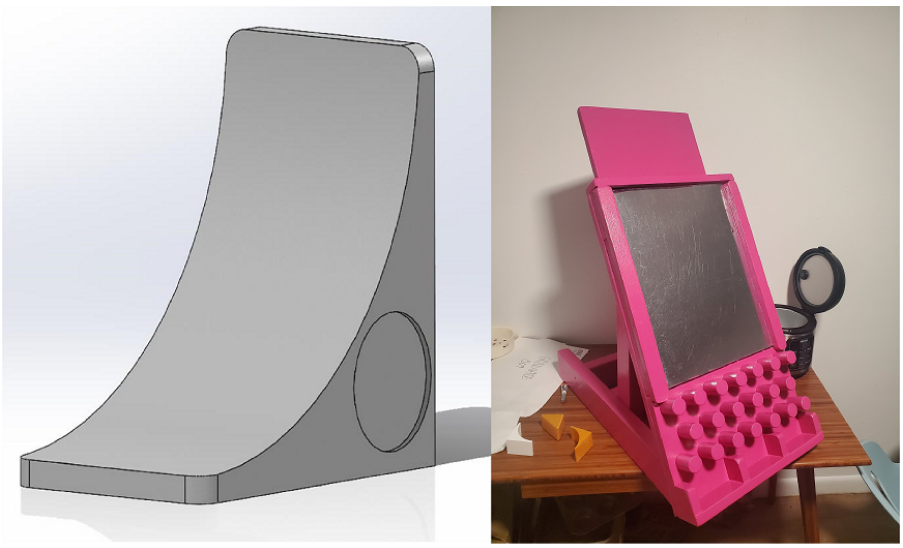


Figure 1. CAD model of the main part. The different subsystems of the main part are shown and pointed towards. The magnetic material used was stainless steel.



(a) (b)

Figure 2. Desing of our teams magnetic marble run exhibit: (a) Beta prototype and (b) SolidWorks desing of a addiditve manufacturable component not to scale. The Beta type included two layers of paint, a metal sheet, and plinko board. The beta type, compared to the earlier alpha prototypes, was the final result of our project. For instance, all of the components were sanded and screwed in for durability and safety. All addiditve manufactured components were plentifully printed.

References

1. S. York, R. Lavi, Y. Dori, and M. Orgil, "Applications of Systems Thinking in STEM Education," *ACS Publications*, May 14, 2019. https://pubs-acs-org.ezaccess.libraries.psu.edu/doi/10.1021/acs.jchemed.9b00261 (accessed Sep. 29, 2023).
2. "Design Project for Discovery Space Museum Exhibit." Discovery Space, State College
3. Aras Ozdemir, Elliot Udell, and Joseph Zelis, "Team 6.7: ME 340 Discovery Space Project," https://youtu.be/OUxkJ41CTpk, 1 December 2023.